NARAYANA
IIT ACADEMY
INDIA

Sec: ISR.IIT _*CO-SC(MODEL-A) Date: 07-04-24
Time: 3 Hrs Max. Marks: 186

07-04-24_ISR.IIT_STAR CO-SC(MODEL-A) _JEE ADV_CAT-24_SYLLABUS

PHYSICS: (100% CUMULATIVE)

SYLLABUS COVERED FROM 23-04-23 TO 06-04-24

CHEMISTRY: (100% CUMULATIVE)

SYLLABUS COVERED FROM 23-04-23 TO 06-04-24

MATHEMATICS: (100% CUMULATIVE)

SYLLABUS COVERED FROM 23-04-23 TO 06-04-24

TIME: 3 HRS **IMPORTANT INSTRUCTIONS** Max Marks: 186

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 8)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-1	8	32
Sec – II (Q.N : 9 – 14)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -1 27.30)	+3	0	6	18
Sec – III (Q.N : 15 – 18)	Matrix Matching Type (Each match consist of 2 questions)	+3	-1	4	12
Total				**18**	**62**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 – 26)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-1	8	32
Sec – II (Q.N : 27 – 32)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+3	0	6	18
Sec – III (Q.N : 33 – 36)	Matrix Matching Type (Each match consist of 2 questions)	+3	-1	4	12
Total				**18**	**62**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 – 44)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-1	8	32
Sec – II (Q.N : 45 – 50)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+3	0	6	18
Sec – III (Q.N : 51 – 54)	Matrix Matching Type (Each match consist of 2 questions)	+3	-1	4	12
Total				**18**	**62**

MATHEMATICS Max.Marks:62

SECTION – I
(MULTIPLE CORRECT CHOICE TYPE)

This section contains **8 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE is/ are correct.**

Marking scheme +4 for correct answer , 0 if not attempted and -1 in all other cases.

1. Let $a_n = 3n + \sqrt{n^2 - 1}$ and $b_n = 2\left(\sqrt{n^2 - n} + \sqrt{n^2 + n}\right), n \ge 1$. If $S_m = \sum_{n=1}^{m} \sqrt{a_n - b_n}$ then

A) $S_{49} + 4S_8 = 13$ B) $S_{49} + 4S_8 = 3$ C) $S_{49}(13 - 4S_8) = 7$ D) $S_{49}(7 - 4S_8) = 13$

2. If the least positive angle satisfying $\sin^3 x + \sin^3 2x + \sin^3 3x = (\sin x + \sin 2x + \sin 3x)^3$ is $\frac{a\pi}{b}$, $a, b \in N$ (Set of naturals). Then

A) Least value of $a + b$ is 12 B) Least value of $a + b$ is 7

C) Least value of $a.b$ is 10 D) Least value of $a.b$ is 35

3. A point M divides A and B in the ratio 1:2 where A and B are diametrically opposite ends of a circle $x^2 + y^2 - 5x - 9y + 22 = 0$. Square AMCD and BMEF on the length AM and MB are constructed on the same side of line AB. If co-ordinates of A in (1,3) then find orthocenter of ΔABE

A) $(1,6)$ B) $(1,5)$ C) $(3,3)$ D) $(4,6)$

4. For the function $f:(0,1) \to R, f(x) = [2^n x] + \{2^m x\}, m, n \in N$ and $n > m$, the number of points of discontinuity of the function can be (where [.], {.} represent greatest integer function and fractional part respectively)

A) 28 B) 26 C) 96 D) 112

5. Let $f(x) = \frac{ax^2 + bx + c}{dx^2 + ex + f}$. If $f(x)$ is not a constant function and both minimum and maximum values of $f(x)$ exist then

A) 4ac must be more than b^2 B) $f(x)$ must be a continuous function

C) 4fd must be more than e^2 D) ae must not be equal to bd

6. Which of the following is/are correct?

A) If A is a $n\times n$ matrix such that $a_{ij}=\left(i^2+j^2-5ij\right)\left(j-i\right)\forall i,j$ then $tr\left(A\right)=0$

B) If A is a $n\times n$ matrix such that $a_{ij}=\left(i^2+j^2-5ij\right)\left(j-i\right)\forall i,j$ then $tr\left(A\right)\neq 0$

C) If P is a 3×3 orthogonal matrix, α,β,γ are the angles made by a straight line with OX, OY, OZ and $A=\begin{bmatrix}\sin^2\alpha & \sin\alpha.\sin\beta & \sin\alpha.\sin\gamma\\ \sin\alpha.\sin\beta & \sin^2\beta & \sin\beta.\sin\gamma\\ \sin\alpha.\sin\gamma & \sin\beta.\sin\gamma & \sin^2\gamma\end{bmatrix}, Q=P^TAP$ then $P.Q^{2025}.P^T=2^{2024}A$

D) If $A=\left[a_{ij}\right]_{3\times 3}$ and $B=\left[b_{ij}\right]_{3\times 3}$ where $a_{ij}+a_{ji}=0$ and $b_{ij}-b_{ji}=0\ \forall i,j$ then $A^{2024}.B^{2025}$ is a singular matrix

7. If $A(1,2,3), B(1,1,1), C(1,1,2)$ and $D(3,-1,2)$ are vertices of a tetrahedron then

A) maximum number of possible planes which is equidistant from all the four vertices A,B,C,D are 4

B) maximum number of possible planes which is equidistant from all the four vertices A,B,C,D are 7

C) If volume of tetrahedron is v and cosine of acute angle between edges AB and CD is ω then $v^2+\omega^2=\frac{19}{90}$

D) If volume of tetrahedron is v and cosine of acute angle between edges AB and CD is ω then $v^2+\omega^2=\frac{104}{25}$

8. Let $\vec{a},\vec{b},\vec{c}$ are three unit vectors such that $\left|\vec{a}-\vec{b}\right|^2+\left|\vec{b}-\vec{c}\right|^2+\left|\vec{c}+\vec{a}\right|^2=3$ then

A) $\frac{3}{2}<\left|\vec{a}+\vec{b}+\vec{c}\right|<\frac{5}{2}$ B) $\left|3\vec{a}+2\vec{b}+\vec{c}\right|>4$ C) $\left|\vec{a}+2\vec{b}+3\vec{c}\right|<5$ D) $\left|\vec{a}+\vec{b}\right|>\left|\vec{b}+\vec{c}\right|$

SECTION – II
(Numerical Value Answer Type)

This section contains 6 questions. The answer to each question is a **Numerical values comprising of positive or negative decimal numbers (place value ranging from Thousands Place to Hundredths place).**
Eg: 1234.56, 123.45, -123.45, -1234.56, -0.12, 0.12 etc.
Marking scheme : +3 for correct answer, 0 in all other cases.

9. If $\sum_{K=1}^{\infty}\frac{1}{K^2}=\frac{\pi^2}{6}, S_n=\sum_{K=1}^{\infty}\frac{n}{\left(36K^2-1\right)^n}$ and $S_1+S_2=\frac{\pi^2}{a}+b$ then $a-b$ is equal to

10. If maximum possible degree of all polynomials whose coefficients are either equal to 1 or – 1 and whose all roots are real is equal to n such that $a^{n-1}+b^{n-1}=7$ and $a^n+b^n=10$ then sum of square of all possible values of $a+b$ is

11. Circle $x^2+y^2-2x-3=0$ intersects parabola $y^2=4x$ at points A and B. Let tangents to parabola at A and B intersect at point C. If radius of circle increases at a rate of 1m/s, the speed of point C after 2025 seconds is (in m/s)

12. The combined equation of the principle axes of the hyperbola $2x^2+12xy-7y^2-16x+2y-5=0$ is $ax^2+2hxy+by^2-x+7y+c=0$ then $|a+b-2h+c|$ is equal to

13. If tangents are drawn at $P_1,P_2,.....,P_n$ on ellipse $\frac{x^2}{64}+\frac{y^2}{9}=1$ intersects the major axis at $T_1,T_2,.....,T_n$ respectively and $\sum_{i=1}^{n}\frac{area\,(\Delta P_iT_iS).area\,(\Delta P_iT_iS')}{(P_iT_i)^2}=18$

(S and S' are foci of ellipse) then n is equal to

14. If two concentric ellipse be such that the foci of one lie on the other and if $\frac{3}{5}$ and $\frac{4}{5}$ be their eccentricities and θ is angle between their axes then $1+\sin^2\theta+\sin^4\theta+\sin^6\theta+-----+\sin^{2024}\theta$ is equal to

SECTION-III
(MATCHING LIST TYPE)

This section contains 4 questions, each having two matching lists (List-1 & List-II). The options for the correct match are provided as (A), (B),(C) and (D) out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer Q.15 and Q.16 by appropriately matching the lists based on the information given in the paragraph.

Let $f(x)=\log_e(a^2-3a-3)|\sin x|+\left[\frac{a^2}{9}\right]\cos(\pi x)\ \forall x\in R$

(Where $[.]$ represents greatest integer function)

LIST – I		LIST –II	
A)	$f(x)$ is periodic with rational fundamental period	P)	$a\in(-\infty,-3]$
B)	$f(x)$ is periodic with irrational fundamental period	Q)	$a\in(-3,-1)\cup\left(-1,\frac{3-\sqrt{21}}{2}\right)$
C)	$f(x)$ is periodic	R)	$a\in\left(-3,\frac{3-\sqrt{21}}{2}\right)$
D)	$f(x)$ is non periodic	S)	$a\in\left(\frac{3+\sqrt{21}}{2},4\right)\cup(4,\infty)$
		T)	$a\in\left(\frac{3+\sqrt{21}}{2},\infty\right)$
		U)	$a\in\{4\}$

15. Which of the following is the only **CORRECT** combination?

A) $B:(Q),(R)$ B) $A:(T),(U)$ C) $B:(Q)$ D) $A:(T)$

16. Which of the following is the only **CORRECT** combination?

A) $D:(T),(U)$ B) $C:(R),(U)$ C) $D:(T)$ D) $C:(S)$

Answer Q.17 and Q.18 by appropriately matching the lists based on the information given in the paragraph.

List I contains conics and List II contains number of common tangents denoted by T and number of common normals denoted by N

	LIST – I		LIST –II
A)	$2y^2=2x-1$ and $2x^2=2y-1$	P)	$T=1$ and $N=1$
B)	$(y-1)^2=\frac{4}{\sqrt{3}}\left(x+\frac{\sqrt{3}}{4}\right)$ and $\frac{\left(x+\frac{\sqrt{3}}{4}\right)^2}{9}+\frac{(y-1)^2}{3}=1$	Q)	$T=2$ and $N=1$
C)	$x^2=y+2$ and $y+x^2-2x+3=0$	R)	$T=1$ and $N=2$
D)	$y^2=-4x$ and $x^2=-4y$	S)	$T=2$ and $N=3$
		T)	$T=3$ and $N=2$
		U)	$T=3$ and $N=1$

17. Which of the following is the only **CORRECT** combination?

A) $B:(Q)$ B) $A:(U)$ C) $B:(T)$ D) $A:(P)$

18. Which of the following is the only **CORRECT** combination?

A) $D:(P)$ B) $C:(R)$ C) $D:(U)$ D) $C:(S)$

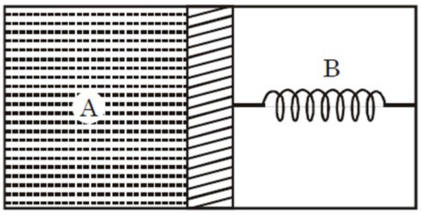
PHYSICS **Max.Marks:62**

SECTION – I
(MULTIPLE CORRECT CHOICE TYPE)

This section contains **8 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE is/ are correct.**

Marking scheme +4 for correct answer , 0 if not attempted and -1 in all other cases.

19. A thermally insulated chamber of volume $2v_0$ is divided by a frictionless piston of area S into two equal parts A and B. Part A has an ideal gas at pressure P_0 and temperature T_0 and part B is vacuum. Initially ideal spring is unstretched. Let in equilibrium the springs be compressed by x_0. Then [spring constant is k]



A) Final pressure of the gas is $\frac{kx_0}{S}$

B) Work done by the gas $\frac{1}{2}kx_0^2$

C) Change in internal energy of the gas is $\frac{1}{2}kx_0^2$

D) Temperature of the gas is decreased.

20. Two bodies A and B have thermal emissivities of 0.01 and 0.81 respectively. The outer surface areas of the two bodies are the same. The two bodies radiate energy at the same rate. The wavelength λ_B, corresponding to the maximum spectral radiancy in the radiation from B, is shifted from the wavelength corresponding to the maximum spectral radiancy in the radiation from A by $1.00\mu m$. If the temperature of A is 5802K,

A) the temperature of B is 1934 K

B) $\lambda_B = 1.5\mu m$

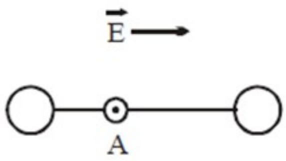
C) the temperature of B is 11604 K

D) the temperature of B is 2901 K

21. A bimetallic strip is formed out of two identical strips one of copper and the other of brass. The coefficient of linear expansion of the two metals are α_C and α_B. On heating, the temperature of the strip goes up by ΔT and the strip bends to form an arc of radius of curvature R. Then R is

A) proportional at ΔT
B) inversely proportional to ΔT
C) proportional to $|\alpha_B - \alpha_C|$
D) inversely proportional to $|\alpha_B - \alpha_C|$

22. There is an insulator rod of length L and of negligible mass with two small balls each of mass m and electric charge Q attached to its ends. The rod can rotate in the horizontal plane around a vertical axis crossing it at a L/4 distance from one of its ends. At first the rod is in unstable equilibrium in a horizontal uniform electric field of field strength E as shown in figure. Then we gently displace it from this position.



A) The maximum velocity attained by the ball which is closer to the axis in the subsequent motion is $\sqrt{\frac{QEL}{5m}}$

B) The maximum velocity attained by the ball which is closer to the axis in the subsequent motion is $\sqrt{\frac{2QEL}{5m}}$

C) When displaced slightly from stable equilibrium position then the time period of the SHM is $2\pi\sqrt{\frac{5mL}{4QE}}$

D) When displaced slightly from stable equilibrium position then the time period of the SHM is $2\pi\sqrt{\frac{2mL}{3QE}}$

23. A double star is a system of two stars of masses m and 2m, rotating about their centre of mass only under their mutual gravitational attraction. If r is the separation between these two stars then their time period of rotation about their centre of mass will be proportional to :

A) $r^{3/2}$
B) r
C) $m^{1/2}$
D) $m^{-1/2}$

24. An ant travels along a long rod with a constant velocity $\vec{u}$ relative to the rod starting from the origin. The rod is kept initially along the positive x-axis. At t = 0, the rod also starts rotating with an angular velocity ω (anticlockwise) in x-y plane about origin. Then

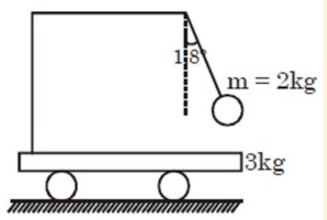
A) the position of the ant at any time t is $\vec{r} = ut\left[\cos\omega t\hat{i} + \sin\omega t\hat{j}\right]$

B) the speed of the ant at any time t is $u\sqrt{1+\omega^2 t^2}$

C) the magnitude of tangential acceleration of the ant at any time t is $\dfrac{\omega^2 tu}{\sqrt{1+\omega^2 t^2}}$

D) the speed of the ant at any time t is $\sqrt{1+2\omega^2 t^2 u}$

25. A simple pendulum of length $\ell = 30\,\text{cm}$ and mass m = 2kg is attached to a cart of mass 3 kg. The cart can move frictionlessly on level ground. The bob is released from rest when string makes an angle of 1.80° with the horizontal:



A) The amplitude of oscillation of cart is $\dfrac{\pi}{5}$cm.

B) The time period of oscillation is $\dfrac{\pi}{5}$sec

C) The cart and bob oscillate with phase difference of π.

D) The time period of oscillation is $\dfrac{\pi\sqrt{3}}{5}$sec.

26. A uniform rod AB of length ℓ and mass M hangs from point A at which it is freely hinged in a car moving with velocity v_0. The rod can rotate in vertical plane about the axis at A. If the car suddenly stops,

A) The angular speed ω with which the rod starts rotating is $\dfrac{3v_0}{2\ell}$

B) The minimum value of v_0 so that the rod completes vertical circular motion is $\sqrt{\dfrac{8}{3}g\ell}$

C) Loss of energy during the process $\dfrac{1}{8}Mv_0^2$

D) There is no loss of energy

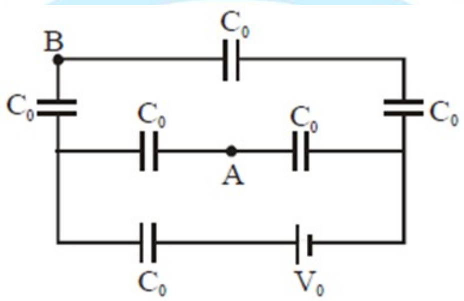
SECTION – II

(Numerical Value Answer Type)

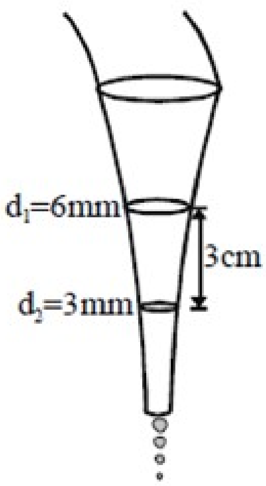
This section contains 6 questions. The answer to each question is a **Numerical values comprising of positive or negative decimal numbers (place value ranging from Thousands Place to Hundredths place).**
Eg: 1234.56, 123.45, -123.45, -1234.56, -0.12, 0.12 etc.
Marking scheme : +3 for correct answer, 0 in all other cases.

27. Two wires made of same material having radii r and 2r are welded together end to end. This combination is used as a sonometer wire kept under tension T. The wedded point is midway between the two bridges.What would be the ratio of the number of loops formed in the wires such that the joint is a node when stationary waves are set up in the wires? If this ratio is (1 : X), then the value of X will be

28. In the arrangement shown in figure, find the potential difference $V_B - V_A$ (in Volt). (Take $V_0 = 55\,V$)



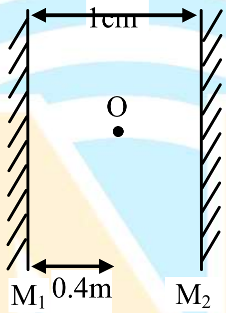
29. The tap in the garden was opened appropriately resulting in the water flowing freely out of it which forms a downward narrowing beam. The beam of water has a circular cross-section, the diameter of the circle is 6 mm at one point and 3 cm below it is only 3 mm as shown in figure. If the rate of water wasted is $(x \times \pi)$ mL/minute then find the value of x. (Neglect the effect of viscosity and surface tension of the flowing water.)



30. A string fixed at both ends is vibrating in the lowest mode of vibration for which a point at quarter of its length from one end is a point of maximum displacement. The frequency of vibration in this mode is 100 Hz. What will be the frequency emitted when it vibrates in the next mode such that this point is again a point of maximum displacement?(in Hz)

31. A glass capillary sealed at the upper end is of length 0.11 m and internal diameter $2\times10^{-5}\,m$. This tube is immersed vertically into a liquid of surface tension $5.06\times10^{-2}\,N/m$. When the length $x\times10^{-2}\,m$ of the tube is immersed in liquid then the liquid level inside and outside the capillary tube becomes the same, then the value of x is : (Assume atmospheric pressure is $1.01\times10^{5}\,\frac{N}{m^2}$)

32. Two plane mirrors are placed parallel to each other at a separation of 1 cm. A point object O is placed at a distance 0.4 cm from mirror M_1 as shown.



Find distance between 5[th] and 6[th] image formed by both mirrors closest to system of mirrors.(in cm)

SECTION-III
(MATCHING LIST TYPE)

This section contains 4 questions, each having two matching lists (List-1 & List-II). The options for the correct match are provided as (A), (B),(C) and (D) out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer Q.33 and Q.34 by appropriately matching the lists based on the information given in the paragraph.

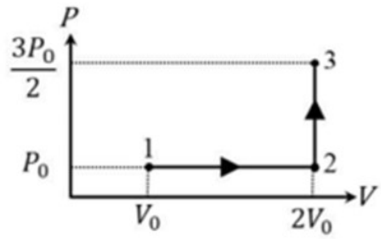
Answer the following by appropriately matching the lists based on the information given in the paragraph.

In a thermodynamic process on an ideal monatomic gas, the infinitesimal heat absorbed by the gas is given by $T\Delta X$, where T is temperature of the system and ΔX is the infinitesimal change in a thermodynamic quantity X of the system. For a mole of monatomic ideal gas $X=\frac{3}{2}R\ln\left(\frac{T}{T_A}\right)+R\,In\left(\frac{V}{V_A}\right)$. Here, R is gas constant, V is volume of gas T_A and V_A are constants.

The List-I below gives some quantities involved in a process and List-II gives some possible values of these quantities.

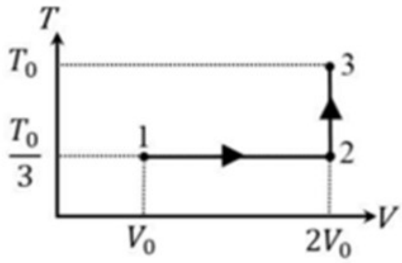
	List-I		List-II
(I)	Work done by the system in process $1 \to 2 \to 3$	(P)	$\frac{1}{3}RT_0 \ln 2$
(II)	Change in internal energy in process $1 \to 2 \to 3$	(Q)	$\frac{1}{3}RT_0$
(III)	Heat absorbed by the system in process $1 \to 2 \to 3$	(R)	RT_0
(IV)	Heat absorbed by the system in process $1 \to 2$	(S)	$\frac{4}{3}RT_0$
		(T)	$\frac{1}{3}RT_0(3+\ln 2)$
		(U)	$\frac{5}{6}RT_0$

33. If the process carried out on one mole of monatomic ideal gas is as shown in figure in the PV-diagram with $P_0V_0 = \frac{1}{3}RT_0$, the correct match is,



A) $I \to Q, II \to R, III \to P, IV \to U$
B) $I \to Q, II \to R, III \to S, IV \to U$
C) $I \to S, II \to R, III \to Q, IV \to T$
D) $I \to Q, II \to S, III \to R, IV \to U$

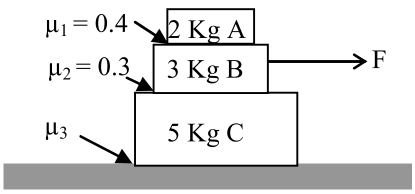
34. If the process on one mole of monatomic ideal gas is an shown in the TV-diagram with $P_0V_0 = \frac{1}{3}RT_0$, the correct match is,



A) $I \to P, II \to R, III \to T, IV \to S$
B) $I \to P, II \to T, III \to Q, IV \to T$
C) $I \to P, II \to R, III \to T, IV \to P$
D) $I \to S, II \to T, III \to Q, IV \to U$

Answer Q.35 and Q.36 by appropriately matching the lists based on the information given in the paragraph.

For the situation shown in the figure given below, Answer the following questions by appropriate match of entries of column I with the entries of column II



	List - I		List - II
I)	If F = 13 N, then	(P)	relative motion between A and B is there
II)	If F = 15 N, then	(Q)	relative motion between B and C is there
III)	If F = 25 N, then	(R)	relative motion between C and the ground is there
IV)	If F = 40 N, then	(S)	relative motion is not there at any of the surface

35. If the value of μ_3 is 0.1 then choose the correct match

(A) I →R ; II →P ; III → R ; IV → P,Q,R

(B) I →R ; II →S ; III →P,Q ; IV → P,R

(C) I →R ; II → R ; III → Q,R ; IV → P,Q,R

(D) I →S ; II →S ; III →P,Q,R ; IV → P,Q,R

36. If the value of μ_3 is 0.3 then choose the correct match

(A) I →S ; II →P ; III → R ; IV → P,Q,R

(B) I →S ; II →S ; III → Q ; IV → P,Q

(C) I →S ; II → R ; III → Q,R ; IV → P,Q

(D) I →S ; II →S ; III →P,Q,R ; IV → P,Q

CHEMISTRY **Max.Marks:62**

SECTION – I
(MULTIPLE CORRECT CHOICE TYPE)

This section contains **8 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE is/ are correct.**

Marking scheme +4 for correct answer , 0 if not attempted and -1 in all other cases.

37. What happens when beryllium oxide is added to baryta water solution?

A) BeO is insoluble in baryta water.

B) BeO dissolves in baryta water.

C) Basic character of the solution increases as the concentration of basic substance increases.

D) pH of the solution decreases.

38. For the reaction:

$A \rightarrow B$

The rate expression is given below:

$$\text{rate} = \frac{2[A]}{3+[A]}$$

Where [A] is the concentration (mol/L) of reactant left at any time $'t\,(\text{in minute})'$ and $[A]_0$ represents initial concentration of the reactant 'A' $(\ln 2 = 0.7)$.

Now, identify the correct statement(s) regarding the above reaction:

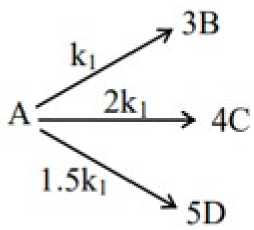
A) Plot of half – life time (Y – axis) versus $[A]_0$ (X – axis) is a straight line with slope 0.25 and intercept 1.05 on y – axis.

B) If initial concentration of A is 8 M, then it takes 5.1 minutes to reduce its concentration to 2 M.

C) The reaction follows first order kinetics if concentration of 'A' is very small as compared to 3.

D) If concentration of A is much greater than 3, then the plot of $t_{1/2}$ (Y – axis) versus $[A]_0$ (X – axis) is a straight line appears to be passing through origin.

39.



All reaction are of 1^{st} order.

At time $t = t_1 (t_1 > 0), \frac{[B]}{[C]} = \alpha$; At time $t = t_2 (\text{where } t_2 \geq t_1), \frac{[C]}{[D]} = \beta$

Which of the following is/are correct?

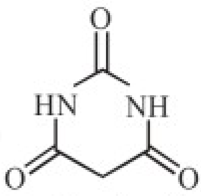
A) $\alpha = 0.5$ B) $\beta = 4/3$ C) $\beta = 0.375$ D) $\beta = 16/15$

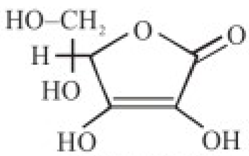
40. Select the correct option(s), mapping column I with column II

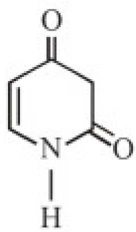
Column – I		Column – II	
(i)	Λm	(a)	Intensive property
(ii)	E^{Θ}_{cell}	(b)	Depends on number of ons/volume
(iii)	κ	(c)	Extensive property
(iv)	$\Delta_r G_{cell}$	(d)	Increases with dilution

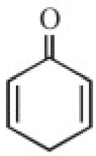
A) $i - a, b, d$ B) $ii - a$ C) $iii - a, b$ D) $iv - b, c$

41. Select the number of compounds in which deprotonation gives aromatic anion:

A) 
Barbituric acid

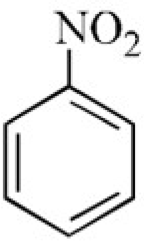
B) 
Ascorbic acid (Vitamin-C)

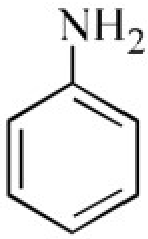
C) 

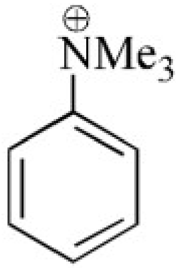
D) 

42. Which of the following does not gives Fridel – Crafts reaction?

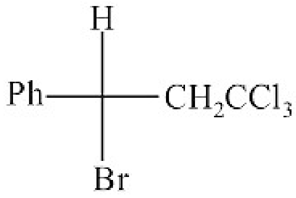
A) 

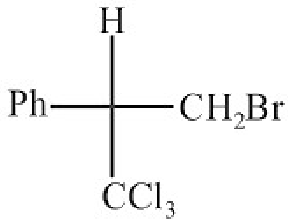
B) 

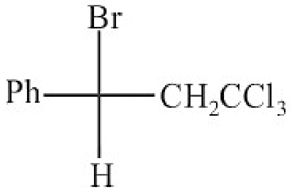
C) 

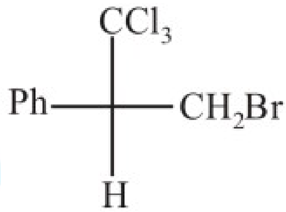
D) 

43. $Ph-CH=CH_2+BrCCl_3 \xrightarrow{\text{Peroxide}} \text{Product is:}$

A) 

B) 

C) 

D) 

44. The ionic addition of HCl to which of the following compounds will produce a compound having Cl on carbon next to terminal.

A) $CF_3.(CH_2)_3.CH=CH_2$

B) $CH_3.CH=CH_2$

C) $CF_3.CH=CH_2$

D) $CH_3.CH_2CH=CH.CH_3$

SECTION – II
(Numerical Value Answer Type)

This section contains 6 questions. The answer to each question is a **Numerical values comprising of positive or negative decimal numbers (place value ranging from Thousands Place to Hundredths place).**
Eg: 1234.56, 123.45, -123.45, -1234.56, -0.12, 0.12 etc.
Marking scheme : +3 for correct answer, 0 in all other cases.

45. The iodine content of a solution was determined by titration with cerium (IV) sulphate in the presence of *HCl*, in which I^- is converted to *ICl*. A 200 mL sample of the solution required 15 mL of $0.05N\ Ce^{4+}$ solution, calculate the iodine concentration in original solution in g/L?

(atomic mass of I = 127g/mol)

46. Let, the maximum number of electrons present in palladium atom $(Z=46)$ satisfying the condition $\ell+m_\ell>0$ be 'x' and the total number of nodal planes in σ^*_{2pz} orbital be 'y', then the value of $\frac{x}{y}$ will be….

47. 6.0 g of He^4 having average velocity $4\times10^2 m/s$ is mixed with 12g of Ne^{20} having same average velocity. If the average kinetic energy per mole of the mixture is 'x' kJ, then find the value of x?

48. A saturated solution of iodine in water is 1.25×10^{-3} M. In 1 L of 0.1 M I^- solution, it is seen that 51.25×10^{-3} mole of I_2 can be maximum dissolved. If the aqueous solution of $I_2(aq)$ undergoes complex formation such as

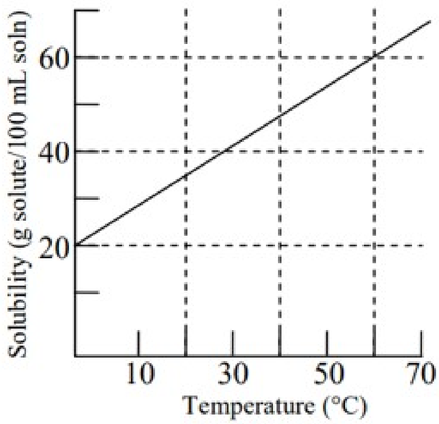
$$I_2(aq) + I^-(aq) \xrightleftharpoons{K_C} I_3^-(aq)$$

What is the value of K_C?

49. How many of the following statements are correct?

1. Enthalpy of formation (with sign) $H_2O < H_2S < H_2Se < H_2Te$
2. Enthalpy of atomization (with sign) $H_2O < H_2S < H_2Se < H_2Te$
3. Dissociation constant (in aq solution) $H_2O > H_2S > H_2Se > H_2Te$
4. Melting point $H_2O < H_2S < H_2Se < H_2Te$
5. Enthalpy of formation (with sign) $NH_3 < PH_3 < AsH_3 < SbH_3$
6. Enthalpy of atomization (with sign) $NH_3 < PH_3 < AsH_3 < SbH_3$
7. Bond angle HXH $NH_3 > PH_3 > AsH_3 > SbH_3$
8. Melting point $NH_3 > PH_3 < AsH_3 < SbH_3$

50. According to the solubility curve shown, how many grams of solute can be recrystallized when 20 mL of a saturated solution at 60ºC are cooled to 0ºC?



SECTION-III
(MATCHING LIST TYPE)

This section contains 4 questions, each having two matching lists (List-1 & List-II). The options for the correct match are provided as (A), (B),(C) and (D) out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer Q.51 and Q.52 by appropriately matching the lists based on the information given in the paragraph.

Consider the following galvanic cell at 25ºC

$$\underset{(1\,bar)}{Pt(s)/H_2(g)}\left|\left(\underset{(100mL,\,0.2M)}{H_3PO_4} + \underset{(100\,mL,0.1M)}{Na_3PO_4}\right)\right|\left|\left(\underset{(100\,mL,0.1M)}{CH_3COOH} \,|\, \underset{(1\,bar)}{H_2\,(g)}/Pt(s)\right)\right.$$

For $H_3PO_4 : K_{a_1} = 10^{-4}, K_{a_2} = 10^{-8}, K_{a_3} = 10^{-12}$ at 25ºC

For $CH_3COOH : K_a = 1\times10^{-5}$ at 25ºC

Now, some electrolyte is added in one of the compartment of the given cell in List – I and corresponding cell potential is given in List – II. Choose the correct option(s) from the codes given below:

$$\left[\text{Take } \frac{2.303RT}{F} = 0.06V, \log 2 = 0.3\right]$$

List – I		List –II	
(I)	If no electrolyte is added in either of the compartment of cell, then	(P)	$E_{cell} = 0.06V$
(II)	If 100 mL, 0.1 M NaOH solution is added in the anode compartment of the original cell, then	(Q)	$E_{cell} = 0.282V$
(III)	If 50 mL, 0.1 M NaOH solution is added in the cathode compartment of original cell, then	(R)	$E_{cell} = 0.18V$
(IV)	If 100 mL, 0.1 M HCl solution is added in the anode compartment of original cell, then	(S)	$E_{cell} = 0.078V$
		(T)	E_{cell} decreases after addition with respect to its initial value.
		(U)	E_{cell} increases after addition with respect to its initial value.

51. Which of the following options has correct combination considering List I and List II?

A) $I \rightarrow R; II \rightarrow S, T;$ B) $I \rightarrow Q; II \rightarrow R, T;$ C) $I \rightarrow R; II \rightarrow Q, U;$ D) $I \rightarrow P; II \rightarrow Q, U;$

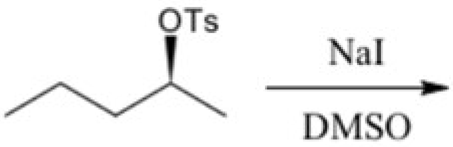
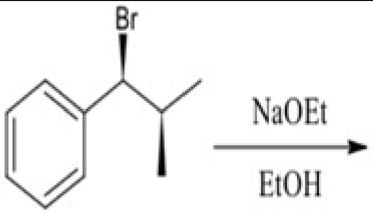
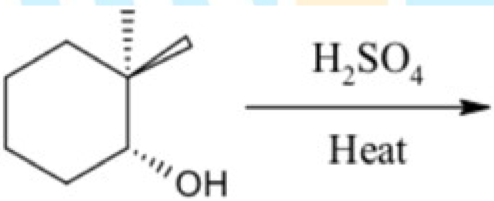
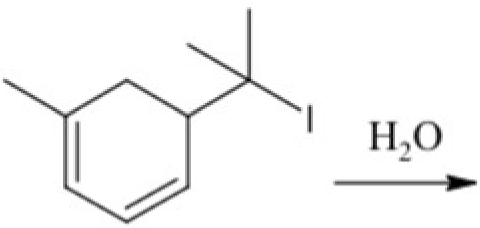
52. Which of the following options has correct combination considering List I and List II?

A) $III \rightarrow P, U; IV \rightarrow S, U$ B) $III \rightarrow Q, U; IV \rightarrow S, T$

C) $III \rightarrow P, T; IV \rightarrow S, T$ D) $III \rightarrow P, T; IV \rightarrow Q, U$

Answer Q.53 and Q.54 by appropriately matching the lists based on the information given in the paragraph.

Match the following column:

	List – I **Reaction**		List – II **(Mechanism involved in formation of major product)**
(P)		(1)	E_1
(Q)		(2)	SN_2
(R)		(3)	SN_1
(S)		(4)	E_2

53. The correct combination is

(A) P2, Q4 (B) P4, Q1 (C) P4, Q4 (D) P4, Q2

54. The correct combination is

(A) R3, S1 (B) R2, S3 (C) R1, S3 (D) R4, S2